PROJECT-1

Title: Railway Gauges

The railway gauges are having three gauges in it, they are:

Broad Gauge: Widest track, used for high-speed and heavy trains; most common in India.

Metre Gauge: Medium width, older system now being converted to broad gauge.

Narrow Gauge: Smallest width, used in hilly or less busy routes

In this project we have to import the csv data to the pandas, we have to analyse it and also visualize the data using matplotlib.

-What are Pandas and use of matplotlib?

Pandas: A Python library used for data analysis and manipulation, especially for handling tables like CSV files.

Matplotlib: A Python library used for data visualization, helping to create graphs like bar charts.

Now, we have to import the csv file to the pandas

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# load CSV file
# df = pd.read_csv('railway_gauges.csv')
df = pd.read_csv("C:/Users/This PC/OneDrive/Desktop/PythonTasks/Analysis/Project/railway_gauges.csv")

print(df.head())
```

The code is a standard template for starting a data analysis project in Python.

Imports: It brings in numpy, pandas, and matplotlib.

Load: It uses Pandas to open a CSV file named railway_gauges.csv

Df.head(): The df.head() command prints the first 5 rows of that table so we can verify the data loaded correctly.

First 5 rows:

We have to understand the data here,

The three gauges data will be displayed including the year and total number of tracks.

#Example

1	Year	Broad Gau	Metre Gau	Narrow Ga	Total
2	1964-65	3521	2891	464	6876
3	1965-66	3603	2913	470	6986
4	1966-67	3615	2911	481	7007
5	1967-68	3601	2920	506	7027

Some of the operations in pandas are:

These are used to access the data

loc[]: Used to select data by labels (names) of rows or columns.

loc → label-based

iloc[]: Used to select data by integer position

iloc → position-based

If suppose we need to find maximum installations, then here is the sample code:

```
# find which year had the max installations
print(df.iloc[[df['Total'].idxmax()]])
```

It will return the data for the year that had the maximum number of installations.

Visualization of the Data:

Data visualization is the graphical representation of information and data, using visual elements like charts, graphs, maps, and dashboards.

Here is the sample code to represent the data,

```
# Plot data using bar chart
df = df.drop('Total', axis=1)
ax = df.plot(x='Year', kind="bar")

plt.xticks(rotation=70)
plt.xlabel('Year')
plt.ylabel('Number of Tracks')
plt.title('Gauges: Number of railway tracks installed per year')

plt.savefig('rail_gauges.png')
plt.show()
```

The provided code performs two main actions:

Data Preparation: It removes the "Total" column from the dataframe to isolate the specific data needed for visualization.

Data Visualization: It generates and saves a labeled bar chart that displays the number of railway tracks installed each year, with rotated x-axis labels for better readability.

What we have done?

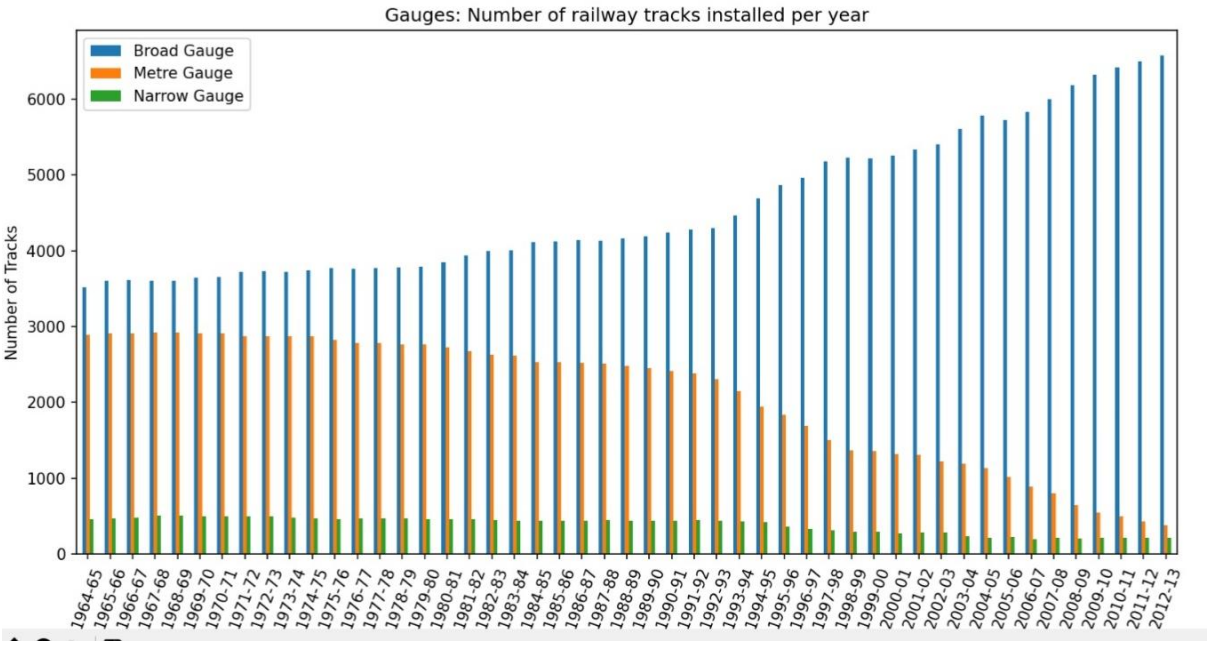
-Imported csv to the pandas

-Identified the year with highest railway track

-Visualization of the data

-Got an idea about how pandas simplify the data.

Output of the process:



Conclusion:

Through the project we have learned about:

Import the file to pandas

Performance of data analysis

Extraction of data and visualization